

Predictive Delinquency Risk Model – Business Summary Report

Prepared for: Head of Collections, Geldium | Data source: 500-customer account dataset (Task 1 EDA)

1. Summary of Predictive Insights

Across the 500-account dataset, 16% of accounts (80 customers) are classified as delinquent. The exploratory analysis highlights several behavioral and financial indicators associated with delinquency risk. Three key risk indicators emerged from the exploratory analysis:

- **Missed payments** – repeated missed payments provide an important behavioral warning signal. Customers with 3 or more missed payments in the last 6 months warrant closer review.
- **Credit utilization** – higher utilization may indicate increased financial pressure. Customers above 70% utilization can be considered for proactive review.
- **Credit score** – lower or declining credit scores may indicate weakening creditworthiness and can support additional risk assessment.

Debt-to-income ratio and account tenure provide additional context, while income should not be used as a stand-alone screening variable. These indicators should be validated through the predictive modelling and pilot evaluation rather than treated as independently proven causal drivers.

Key Insight	Customer Segment	Influencing Variables	Potential Impact
Repeated missed payments are an important early-warning signal	Accounts with ≥ 3 missed payments in 6 months	Missed_Payments, Month_1–Month_6 status	Early outreach before the account becomes delinquent
High utilization may indicate increased financial pressure	Accounts with utilization > 70%	Credit_Utilization, Loan_Balance	Proactive credit-limit or hardship conversation
Low / falling credit score signals weakening creditworthiness	Score < 450 or a 50+ point drop	Credit_Score	Trigger manual account review

Top 3 Risk Factors: (1) missed payments, (2) high credit utilization, (3) low/declining credit score.

2. Recommendation Framework

Restated Insight: Customers with 3+ missed payments in the last 6 months and credit utilization above 70% are the clearest early-warning combination for delinquency risk.

Proposed Recommendation: Launch a controlled proactive outreach and support pilot for customers identified as higher risk using the model outputs and validated EDA-based indicators, offering payment reminders and a fast path to short-term hardship or debt-management support.

Specific

Use the predictive model's selected probability threshold to identify customers for the pilot. Where model confidence is limited, supplement the review with EDA-based indicators such as repeated missed payments, high credit utilization, and low credit score. High-impact actions remain subject to human review.

Measurable

Track 30/60/90-day delinquency rate and hardship-plan enrollment for the pilot group versus a matched control group; target a 10–15% relative reduction in new delinquencies among flagged customers.

Actionable

Uses fields already in the data (Missed_Payments, Credit_Utilization, Credit_Score) and existing Collections contact channels – only threshold rules from the model need to be applied to the outreach queue.

Relevant

Targets key risk indicators identified during the EDA and provides a controlled way to test whether these indicators can improve early collections intervention.

Time-bound

6-week pilot, with a go/no-go review at week 6 for portfolio-wide rollout.

Justification and Business Rationale

This action is scalable (rules-based flagging extends to the full portfolio once validated), easy to implement (uses existing Collections workflows), fair (reviewed via the fairness audit across Location, Employment_Status, Age and Card_Type so outreach isn't skewed toward any one group), and cost-effective (early, supportive outreach is materially cheaper than post-delinquency collections). It is grounded in the exploratory analysis, which identified payment behavior and credit utilization as important risk indicators. Because the predictive models showed limited performance on the supplied dataset, the recommendation is structured as a controlled pilot rather than an immediate portfolio-wide deployment.

3. Ethical and Responsible AI Considerations

Fairness risks and mitigations

- **Missing-data bias** – Income (7.8% missing) and Loan_Balance (5.8% missing) could be imputed inaccurately for specific segments (e.g., self-employed customers). Mitigation: use group-based/KNN imputation and check whether missingness itself correlates with a segment before trusting imputed values.
- **Proxy-variable bias** – Location (5 cities) or Credit_Card_Type could stand in for income or socioeconomic status. Mitigation: run the model's Fairness Audit step to compare error rates across these groups, and use SHAP to confirm predictions rely on behavior (missed payments, utilization) rather than these proxies.

Explainability

SHAP-based explanations can be incorporated, where applicable to the selected model, so the Collections team can understand which factors contributed to an individual customer's risk score in plain language.

Responsible decision-making

The recommended action is supportive (outreach and hardship options) rather than punitive (e.g., automatic credit-limit cuts), and Moderate/High-risk cases are routed to human review rather than triggering automated account actions. Customer financial data is protected throughout, and the model should be re-audited periodically as customer behavior and economic conditions change.